

# Programming for Data Analytics II: Session 6

## ETL Pipeline with Pandas

# Overview

A simple ETL (Extract, Transform, Load) pipeline using Pandas and CSV files.

# Pipeline Architecture

1. **Sources / Raw:** Original data files (CSV).
2. **Staging:** Cleaned data, standardized types.
3. **Intermediate:** Enriched data, joins, business logic.
4. **Marts:** Aggregated data ready for reporting.

# 1. Sources / Raw

- **Location:** `data/raw/`
- **Files:**
  - `customers.csv` : Customer details.
  - `products.csv` : Product catalog.
  - `orders.csv` : Transactional data.
- **State:** Raw, potentially messy, untyped.

## 2. Staging

- **Location:** `data/staging/`
- **Process:**
  - Read raw CSVs.
  - Convert dates to datetime objects.
  - Standardize column names (if needed).
  - Handle basic data quality issues.
- **Goal:** Prepare data for joining.

### 3. Intermediate / Business Logic

- **Location:** `data/intermediate/`
- **Process:**
  - Join `orders` with `customers` and `products` .
  - Calculate derived columns (e.g., `total_amount = quantity * price` ).
  - Filter invalid records.
- **Goal:** Create a "Wide Table" or "Enriched" dataset containing all necessary context.

## 4. Marts

- **Location:** data/marts/
- **Process:**
  - Group by dimensions (Category, Date, Country).
  - Aggregate metrics (Sum of Revenue, Count of Orders).
- **Goal:** Small, focused tables for specific reports or dashboards.

## Summary

This pipeline demonstrates how to structure data processing into logical layers, making the code more maintainable, debuggable, and scalable.